

Evaluating Hallucination Detectors for RAG Systems

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Abstract

Detecting answers that are not supported by the retrieved context is a prerequisite for monitoring retrieval-augmented generation (RAG) systems. We compare four faithfulness detectors on HaluBench while holding the judge model fixed (gpt-4o-mini), so that differences reflect the evaluation protocol rather than model capability: a holistic LLM judge, a claim-level LLM judge (LLM-Claims), G-Eval, and LLM claim extraction followed by natural language inference (NLI). On a stratified sample of 1,198 examples from six subsets, the three LLM-based detectors are statistically indistinguishable (accuracy 79.6–82.2%, ROC-AUC 83.9–84.9%), while the NLI pipeline is far weaker (accuracy 59.2%) because it labels correct paraphrases as unsupported. On RAGTruth, the only subset with naturally occurring hallucinations, LLM-Claims ranked hallucinations best but its decision rule missed them. In a pre-registered follow-up on 700 held-out RAGTruth answers, flagging an answer when *any* claim fails raised recall from 36.3% for the holistic judge’s verdict to 76.6% at similar precision (50.0% vs. 48.4%); even with its threshold tuned post hoc, the holistic judge reached at most 46.8% recall at 50% precision. All three pre-registered hypotheses were supported with paired bootstrap confidence intervals excluding zero. These findings are limited to one dataset of natural hallucinations (124 positive examples) and one judge model.

1 Introduction

A RAG system can retrieve the right passage and still generate an answer that adds or alters facts. Because human review of every answer is not feasible in production, automatic faithfulness detection is needed to monitor such systems.

Using a large language model as a judge is now the common approach, but a single model can serve as a judge in several ways: it can be asked for one verdict on the whole answer [4], asked to judge each atomic claim separately, following the decomposition idea of FActScore [3], or asked for a graded score on a rubric [2]. Cheaper non-LLM alternatives based on natural language inference are also used, for example in RAGAS [1]. Published comparisons usually change the model and the protocol at the same time, which makes it hard to tell how much the protocol itself matters.

This report holds the judge model fixed and addresses three research questions:

- RQ1.** With the same model, do holistic, claim-level and rubric-based judging differ in how well they detect hallucinations?
- RQ2.** How does a claim-extraction + NLI pipeline compare with LLM judges?
- RQ3.** Does the answer depend on the kind of hallucination: synthetic perturbations versus hallucinations that occur naturally in RAG outputs?

Contributions. (i) A controlled comparison of four detectors on six HaluBench subsets, with bootstrap confidence intervals and paired tests; (ii) an analysis of why NLI-based checking fails on RAG answers; (iii) a pre-registered, held-out test showing that on RAGTruth’s naturally occurring

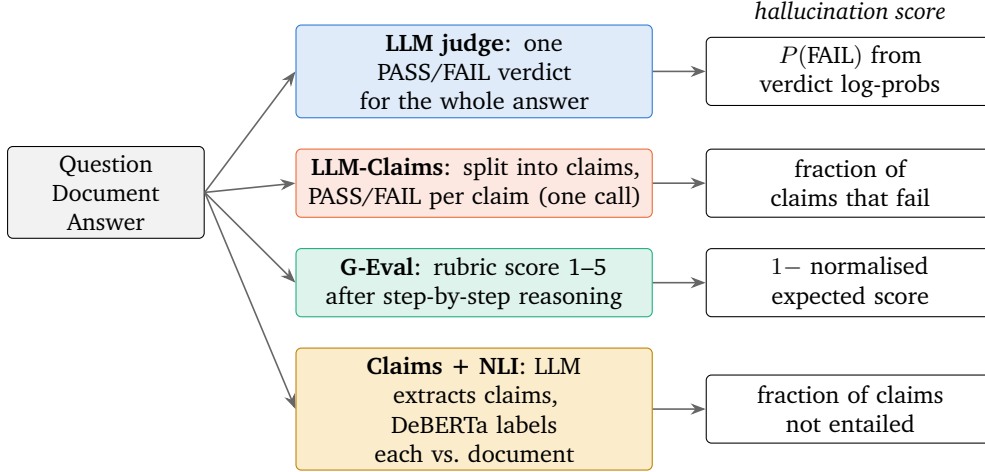


Figure 1: The four evaluation protocols. All LLM steps use gpt-4o-mini at temperature 0, so the protocols differ only in how the question is posed and how the output becomes a score.

Table 1: Detector details. Decision rules were fixed before any evaluation and were not tuned on the data.

Detector	Decision rule (main run)	Calls / answer	Based on
LLM judge	The model’s own PASS/FAIL verdict	1	HaluBench / Lynx prompt [4]
LLM-Claims	Hallucinated if $> 50\%$ of claims fail	1	Claim decomposition [3]
G-Eval	Hallucinated if score > 0.5	1	Liu et al. [2]
Claims + NLI	Any contradiction, or $> 60\%$ of claims neutral	1 + local model	e.g. RAGAS [1]

hallucinations a claim-level judge with an “any failing claim” rule substantially improves recall at similar precision; and (iv) an open, resumable evaluation harness with 29 offline tests.

2 Detectors

All four detectors receive the same (question, document, answer) triple. Each returns a binary decision (hallucinated or not) and a continuous hallucination score in $[0, 1]$, so both thresholded and threshold-free metrics can be computed. Figure 1 summarises the protocols and Table 1 their details.

For the NLI detector, the cross-encoder `cross-encoder/nli-deberta-v3-large` accepts roughly 512 tokens. Documents are therefore split into overlapping 400-token windows, each paired with the claim as a proper (premise, hypothesis) pair; a claim counts as entailed if any window entails it. This avoids truncating the claim itself when the document is long.

3 Experimental setup

Data. HaluBench [4] contains about 15,000 (question, document, answer) triples from six sources, each labelled PASS (faithful) or FAIL (hallucinated). In five subsets the hallucinated answers are constructed: generated by an LLM (HaluEval) or perturbed from correct answers (FinanceBench, DROP, CovidQA, PubMedQA). In RAGTruth they are real outputs of RAG systems annotated by humans (Table 2).

Sampling. Examples are drawn with a fixed seed and stratified by label, keeping each subset’s own label ratio. This matters because HaluBench is ordered by label: taking the first N rows, as

Table 2: Evaluation data. The main run draws 200 examples per subset; the held-out run uses every RAGTruth example not used in the main run.

Subset	Domain	Hallucinated answers	Main: n (FAIL)	Held-out: n (FAIL)
HaluEval	General QA	LLM-generated	200 (100)	–
RAGTruth	Mixed RAG tasks	Natural, human-labelled	200 (36)	700 (124)
FinanceBench	Financial filings	Perturbed	200 (100)	–
DROP	Wikipedia, numerical	Perturbed	200 (100)	–
CovidQA	COVID-19 research papers	Perturbed	200 (100)	–
PubMedQA	Biomedical abstracts	Perturbed	200 (100)	–

Table 3: Pooled results on 1,198 HaluBench examples (% , 95% bootstrap CI). Δ is the paired difference in accuracy to the LLM judge. Cost is API cost per 1,000 answers at \$0.15/\$0.60 per million input/output tokens.

Detector	Accuracy	Bal. acc.	F1	ROC-AUC	PR-AUC	Δ accuracy	USD
LLM judge	82.2 [79.9, 84.5]	82.0	80.0	83.9 [81.4, 86.2]	78.0	–	0.24
LLM-Claims	80.8 [78.5, 83.1]	80.0	77.1	84.9 [82.6, 86.9]	77.0	–1.4 [–3.6, +0.8]	0.30
G-Eval	79.6 [77.4, 81.9]	79.2	76.7	84.4 [82.1, 86.6]	78.0	–2.6 [–4.8, –0.4]	0.26
Claims + NLI	59.2 [56.4, 61.7]	59.9	59.4	61.5 [58.4, 64.1]	54.0	–23.0 [–26.3, –19.9]	0.06 ^a

^a Plus local inference of the NLI model (about 3 hours for 1,200 answers on Apple silicon).

an earlier version of this project did, returns a single class for several subsets and makes precision, recall and F1 meaningless.

Metrics. The positive class is “hallucinated”. We report accuracy, balanced accuracy, precision, recall and F1 for binary decisions, and ROC-AUC and PR-AUC for continuous scores. RAGTruth is imbalanced (18% hallucinated), so always answering “faithful” already reaches 82% accuracy; balanced accuracy, F1 and PR-AUC are the informative metrics there.

Statistics. 95% confidence intervals are percentile bootstrap intervals over examples (1,000 resamples in the main run, 2,000 in the held-out run). Differences between detectors use a paired bootstrap on the same examples; a difference is considered established only when its interval excludes zero.

Error handling. A failed API call or unparseable reply is recorded as an error, never as “faithful”. Two of 1,200 main-run examples still failed after retries (one unparseable LLM-Claims reply, one failed claim extraction for NLI); they are excluded for every detector so that all detectors are compared on the same examples. The held-out run had no errors.

4 Results on HaluBench

4.1 Overall comparison

Table 3 shows that the three LLM-based detectors perform almost identically, while claims + NLI is 23 accuracy points behind (RQ1, RQ2).

The ROC-AUCs of the three LLM detectors (83.9–84.9) have overlapping intervals. LLM-Claims’ accuracy gap to the judge is not distinguishable from zero. G-Eval is slightly less accurate, but its interval only just excludes zero and it is one of three comparisons, so this is weak evidence. With gpt-4o-mini, neither claim decomposition nor log-probability weighting improves on a single holistic verdict across HaluBench as a whole.

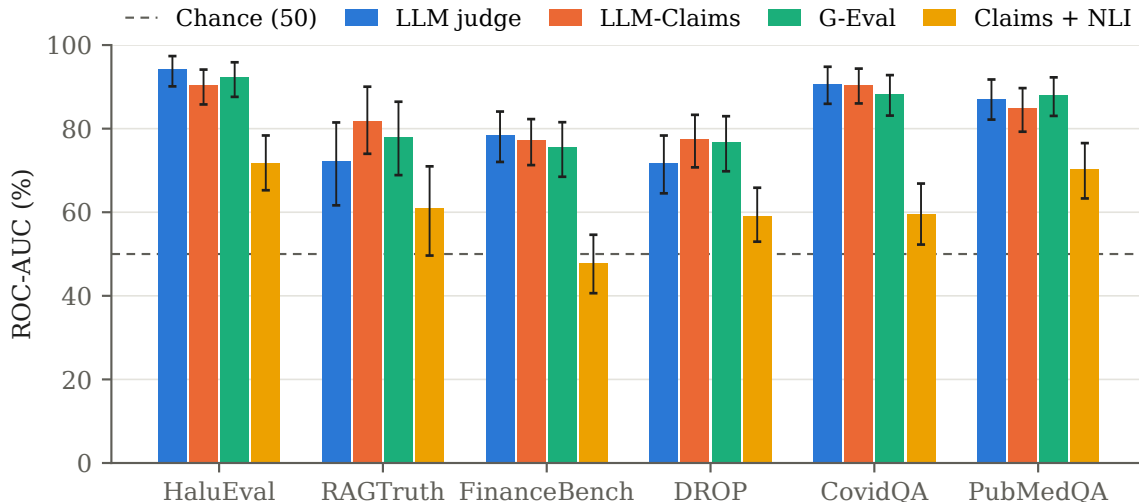


Figure 2: ROC-AUC per HaluBench subset with 95% bootstrap intervals (200 examples per subset; 199 for HaluEval and FinanceBench after error exclusion). The three LLM detectors are close on every subset; NLI is near chance on FinanceBench.

For context, the holistic judge’s macro-averaged accuracy (82.2%) lies in the range reported by Ravi et al. [4] for Lynx-8B (82.9%) and Llama-3-70B (80.1%), below GPT-4o (86.5%) and above RAGAS faithfulness (66.9%). Their figures cover the full benchmark and other models, so this comparison is indicative only.

4.2 Per-subset results

Figure 2 shows that the LLM detectors track each other closely on most subsets. The exceptions are RAGTruth and DROP, where LLM-Claims ranks hallucinations better than the holistic judge (81.8 vs. 72.1 and 77.4 vs. 71.7), although per-subset intervals overlap at this sample size. HaluEval is the only subset where the holistic judge leads on accuracy (92.5% vs. 85.9% for LLM-Claims), with barely overlapping intervals.

4.3 Why NLI fails

Claims + NLI flags 47% of faithful answers as hallucinated (312 of 663) and still misses 33% of hallucinations (178 of 536). Of the 312 false alarms, 195 stem from correct claims labelled *neutral*: the NLI model does not treat paraphrases, summaries, or claims that combine two sentences as entailed. The remaining 117 stem from spurious *contradiction* labels. On FinanceBench, where answers depend on numbers from tables, its ROC-AUC (47.7) is at chance.

An earlier version of this pipeline reported 76% accuracy for NLI. That figure was an artifact of the sampling problem described in Section 3: on subsets where every sampled answer was hallucinated, a detector that over-flags appears accurate.

4.4 A pattern on RAGTruth

On RAGTruth, accuracy is uninformative: all LLM detectors score 81–83%, the same as always answering “faithful”. LLM-Claims had the best ranking (ROC-AUC 81.8) but the worst F1 (14.6), because its rule “more than 50% of claims fail” let almost every hallucination through. Only 3 of the 36 hallucinated answers crossed 50%, whereas 27 of 36 had at least one failing claim (compared with 25 of 164 faithful answers).

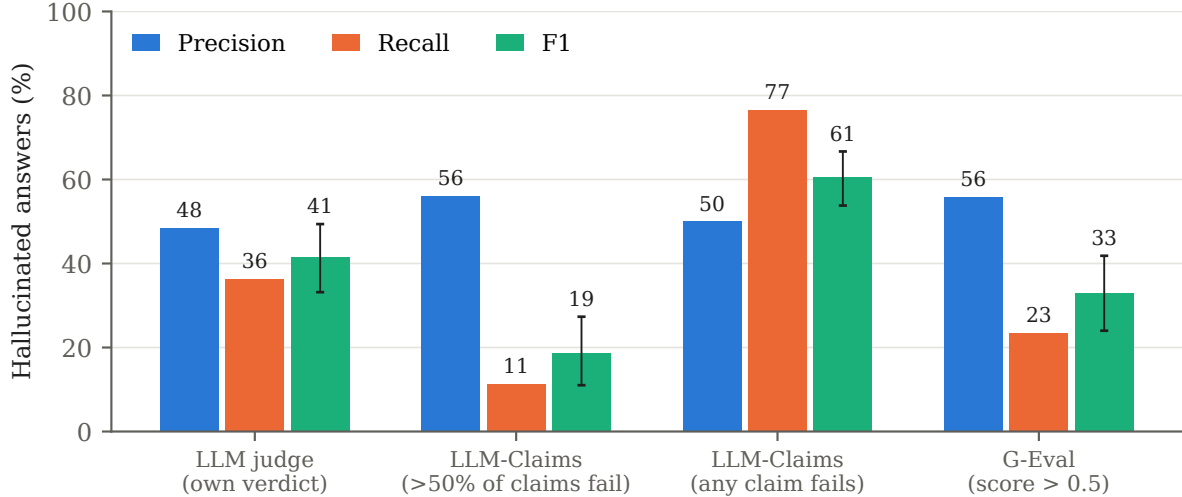


Figure 3: Precision, recall and F1 per decision rule on 700 held-out RAGTruth answers (124 hallucinated). Whiskers on F1 show 95% bootstrap intervals. Flagging an answer when any claim fails roughly doubles recall relative to the judge’s own verdict, at similar precision.

Table 4: Held-out RAGTruth results (% , 95% bootstrap CI; $n = 700$, 124 hallucinated). The two LLM-Claims rows share one set of scores, so their ROC-AUC and PR-AUC are identical by construction. Accuracy is omitted: every rule scores 82–83%, the same as always answering “faithful”.

Decision rule	Prec.	Recall	F1	Bal. acc.	ROC-AUC	PR-AUC
LLM judge (own verdict)	48.4	36.3	41.5 [33.2, 49.4]	64.0 [59.7, 68.4]	73.2 [67.7, 78.4]	40.5
LLM-Claims (>50% of claims fail)	56.0	11.3	18.8 [11.0, 27.3]	54.7 [52.2, 57.6]	81.1 [76.8, 85.3]	46.9
LLM-Claims (any claim fails)	50.0	76.6	60.5 [53.8, 66.7]	80.1 [75.9, 84.0]	81.1 [76.8, 85.3]	46.9
G-Eval (score > 0.5)	55.8	23.4	33.0 [24.0, 41.8]	59.7 [56.1, 63.6]	76.0 [70.9, 80.5]	39.2

This suggests that natural hallucinations are usually a minority of wrong claims inside otherwise correct answers. Because the pattern was found on the evaluation data itself, it was treated as a hypothesis and tested on new data.

5 Pre-registered held-out test on RAGTruth

5.1 Design

Before any held-out example was scored, the following hypotheses, decision rules, data and analysis were written down (`docs/ragtruth_holdout.md` in the repository):

- **H1** (replication, threshold-free): LLM-Claims ranks hallucinations better than the holistic judge (ROC-AUC).
- **H2** (decision rule): “any claim fails” outperforms “more than 50% of claims fail” (F1).
- **H3** (practical): LLM-Claims with the “any” rule outperforms the holistic judge (F1).

The held-out set comprises every RAGTruth example not used in the main run: 700 answers, 124 hallucinated. Detectors, prompts and model are unchanged; only the LLM-Claims threshold differs, and both rules are computed from the same saved scores. A hypothesis is supported if the 95% paired bootstrap interval of the difference excludes zero in the stated direction.

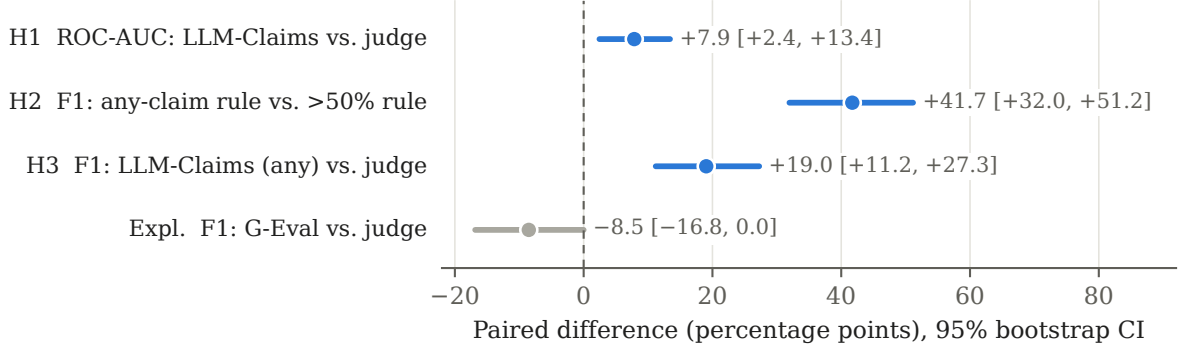


Figure 4: Paired differences for the pre-registered hypotheses (blue) and one exploratory comparison (grey), in percentage points with 95% bootstrap intervals (2,000 resamples). All three pre-registered intervals exclude zero.

Table 5: Hypothesis tests on the held-out set (paired bootstrap, percentage points).

	Comparison	Δ	95% CI	Outcome
H1	ROC-AUC: LLM-Claims vs. judge	+7.9	[+2.4, +13.4]	Supported
H2	F1: any-claim rule vs. >50% rule	+41.7	[+32.0, +51.2]	Supported
H3	F1: LLM-Claims (any) vs. judge	+19.0	[+11.2, +27.3]	Supported
Expl.	F1: G-Eval vs. judge	-8.5	[-16.8, 0.0]	Exploratory

5.2 Results

As Figure 3 and Table 4 show, the holistic judge catches 36% of hallucinated answers, whereas LLM-Claims with the “any” rule catches 77%. Both are correct about half the time when they raise a flag (precision 48% and 50%). The original “more than 50%” rule is the most precise but misses 89% of hallucinations.

All three hypotheses are supported (Figure 4, Table 5). Balanced accuracy, a secondary outcome, agrees with H3 (+16.1 points, CI [11.4, 20.8]). The G-Eval comparison was exploratory; its interval touches zero.

5.3 Is the comparison fair to the judge?

Recall at a single operating point favours whichever detector has the better-chosen threshold. The judge also outputs a continuous score, $P(\text{FAIL})$, so a lower cut-off could buy it more recall. In an exploratory analysis (not pre-registered) we therefore swept the judge’s threshold over all observed values on the held-out set. Its scores are almost binary (the median is 0.0 and the 90th percentile 1.0), so lowering the cut-off helps little: at precision $\geq 50\%$ its best recall is 46.8%, and at precision $\geq 45\%$ it is 55.6%. These are best cases, tuned on the test data itself; LLM-Claims with its pre-declared rule reaches 76.6% recall at 50.0% precision. The advantage is therefore not an artifact of the judge’s default threshold, but its size is closer to $1.6\times$ than $2\times$ against a tuned judge. The threshold-free comparison (H1, ROC-AUC +7.9 points) is the fairest single summary.

5.4 Consistency with the development sample

The held-out estimates closely reproduce the numbers first observed on the 200 main-run RAGTruth answers: F1 with the “any” rule $61.4 \rightarrow 60.5$, LLM-Claims ROC-AUC $81.8 \rightarrow 81.1$, and the judge’s ROC-AUC $72.1 \rightarrow 73.2$. The detectors behave consistently on new data, and the effect is not an artifact of one sample.

6 Discussion

In our experiments the evaluation protocol mattered little on the synthetic subsets and substantially on RAGTruth, the one subset with natural hallucinations (RQ3). Whether this holds for natural hallucinations in general needs more datasets to establish.

Synthetic subsets. In five subsets the hallucinated answer is a deliberate perturbation of a correct one: a changed number, a swapped entity, an added unsupported statement. A likely explanation for the tie is that the error is then prominent, so a holistic verdict and a claim-by-claim check detect the same thing.

Natural hallucinations. Natural hallucinations appear to be mostly one or a few unsupported details inside an otherwise correct, often long answer. Asked for one overall verdict, the judge weighs many correct statements against one wrong one and often answers PASS; asked about each claim separately, it has to rule on the wrong detail on its own. The per-claim scores are consistent with this (Section 4.4).

The decision rule is part of the method. The same LLM-Claims scores yield an F1 of 18.8 or 60.5 depending only on the threshold. Ranking ability (ROC-AUC) and usefulness at a fixed cut-off can diverge sharply, which is why both are reported and why the rule was fixed before the held-out test.

NLI is not a cheap substitute. Off-the-shelf NLI cross-encoders are trained on short, literal premise–hypothesis pairs, whereas RAG answers paraphrase and summarise. The pipeline is cheaper in API cost but loses 23 accuracy points, which makes it unsuitable as a standalone detector here.

Practical recommendations.

- To monitor a RAG system, use a claim-level LLM judge and flag an answer when any claim fails. It costs about $1.9\times$ a holistic judge (\$0.32 vs. \$0.17 per 1,000 answers on RAGTruth with gpt-4o-mini) and, in our RAGTruth experiments, catches substantially more hallucinations (76.6% vs. 36.3% for the judge’s verdict, or at most 46.8% with a tuned threshold). Validate this on your own data before relying on it.
- At 50% precision, half of the flags are false alarms: use the detector to route answers to review, not to block them automatically.
- Evaluate detectors on naturally occurring hallucinations and report precision and recall rather than accuracy, which hides a detector that misses most hallucinations on imbalanced data.
- Keep the per-claim verdicts: they show which sentence failed, which speeds up review and error analysis.

7 Limitations

- **One judge model.** All LLM steps used gpt-4o-mini. A stronger judge may catch single-claim errors in a holistic verdict and narrow the gap.
- **One prompt per detector.** LLM judges are sensitive to wording; the results hold for the prompts in the repository.

- **Generalisation of the main finding.** The claim-level advantage was shown on one dataset of natural hallucinations (RAGTruth, 124 positive examples in the held-out set), with one model and one prompt. It should be treated as a well-supported result for that setting, not as a general law for RAG systems.
- **Scope of the “any” rule.** It was tested only on RAGTruth; on the synthetic subsets it may raise more false alarms.
- **Sample size.** 200 examples per subset give per-subset intervals of roughly ± 5 accuracy points; only large per-subset differences are resolved.
- **Published baselines.** The Lynx figures cover the full benchmark and other models; they are context, not a head-to-head comparison.
- **Synthetic data.** Five of six subsets use constructed hallucinations, which may be easier to detect than those produced by deployed systems.
- **Pre-registration record.** The held-out hypotheses were written before the held-out run, but were not committed to version control beforehand, so no independent timestamp exists.

8 Reproducibility

All experiments run from the accompanying repository with one command each and cost about \$1.60 in API calls in total (main run \$1.03, held-out run \$0.46). Sampling is seeded and the exact example identifiers of each run are stored; LLM calls use temperature 0, which is near-deterministic but not bit-identical across API versions. Predictions are appended per example, so interrupted runs resume. The figures and results tables in this report are generated directly from the saved result files by `report/make_figures.py`.

```
python run_benchmark.py --out-dir results/main
python run_benchmark.py --subsets RAGTruth --n 0 --exclude-from results/main \
    --methods llm_judge llm_claims geval --claims-threshold 0 \
    --out-dir results/ragtruth_holdout
python run_benchmark.py --out-dir results/ragtruth_holdout --report-only \
    --bootstrap 2000
python report/make_figures.py
```

References

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